CatchPoint

Let them step. Catch only when they can’t.



ME5412 project option · working draft · 22 Oct 2026

Concept render (design D3, unbuilt): Blender; mannequin Mixamo/Adobe via three.js examples; HDRI Poly Haven (CC0).

Recovery steps prevent falls — but practising them is risky



71%

of stroke patients still had impaired reactive stepping at discharge

Inness et al., Phys Ther 2014

14.7%

fell during inpatient stroke rehabilitation

Mansfield et al., Neurorehabil Neural Repair 2013

684,000

deaths from falls every year worldwide

World Health Organization

Photo: US Navy, public domain (Wikimedia Commons). Today therapists guard patients by hand, so they tend to over-support.

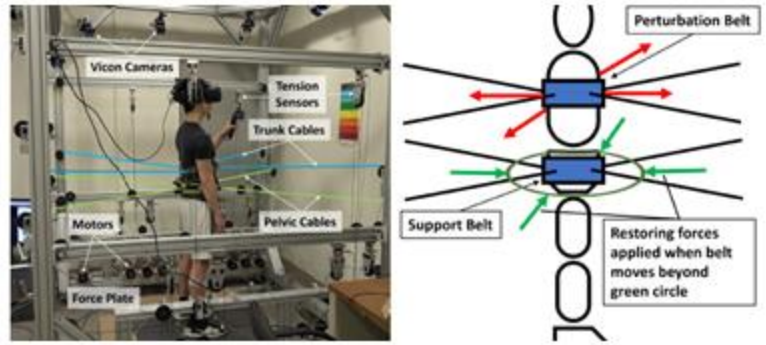
Today’s robots lock too early or too late



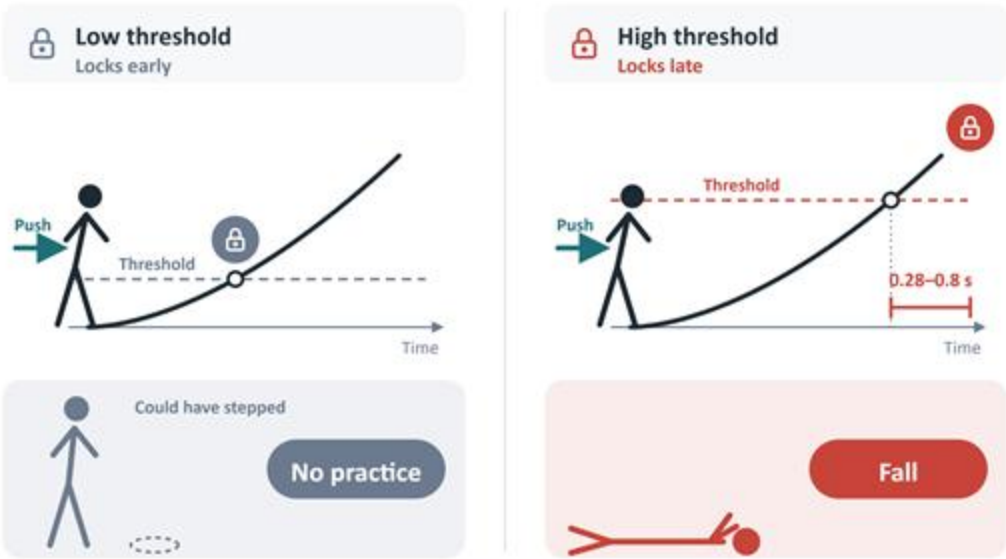
MRBA (NTU): brakes lock when a sensor threshold trips

Photo omitted in public version
(Hocoma Andago — see source link)

Andago (Hocoma): overground support walker



RobUST: pelvic force field outside a fixed boundary



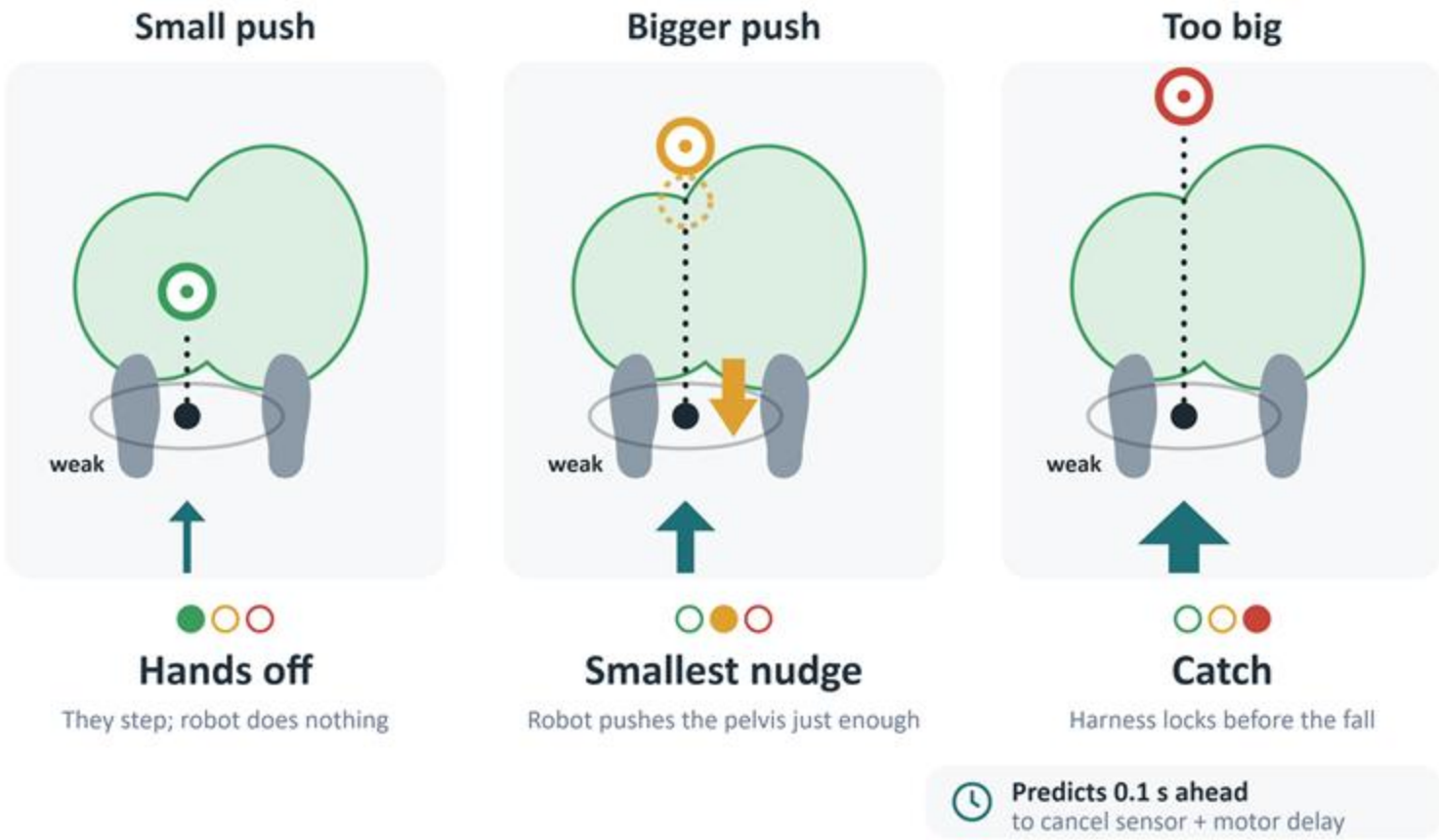
Locks early — the patient never gets to practise stepping.

Locks late — response takes 0.28–0.8 s — they fall anyway.

Photos: NTU MRBA, J NeuroEng Rehabil 2023 (CC BY 4.0) · Hocoma Andago, © Hocoma/DIH · RobUST, Omofuma et al., Bioengineering 2023 (CC BY). Lock delays: MRBA 0.28 s, IMU walker 0.8 s.

Our idea: help only when a step can no longer save them

Ask "can they still save themselves?" — not "have they started to fall?"



How to read it

◎ = where they must step to stay upright (capture point)

Green area = where this patient's own step can reach

Inside it: hands off. Outside: smallest nudge. Out of reach: catch.

Illustration: team figure (schematic, top view). The capture point is a standard balance concept (Pratt 2006; Hof 2005).

Same push, three responses

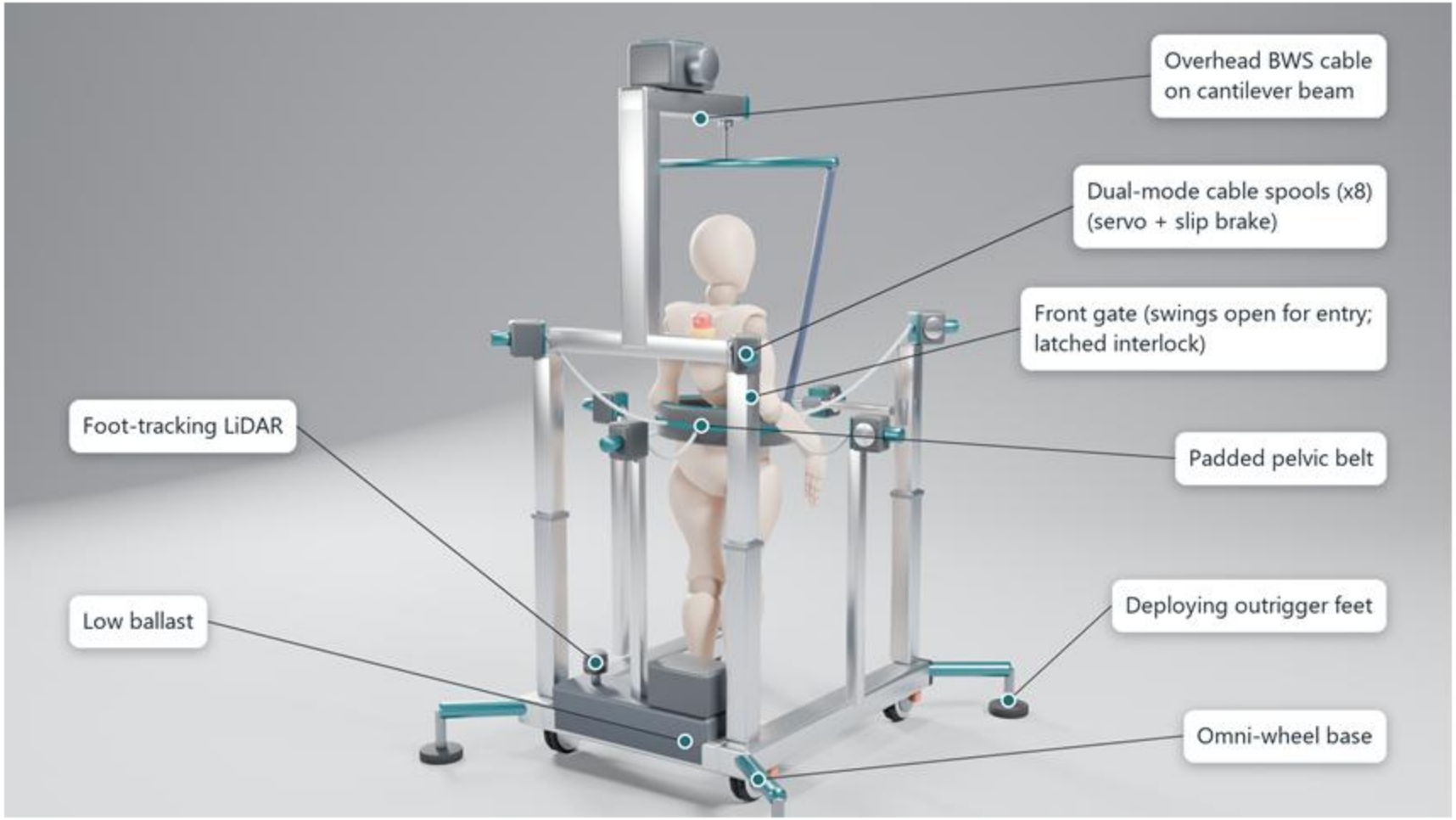


One cable set, three modes: slack · graded nudge ≤ 150 N · passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total

Design limits. Concept render of design D3: unbuilt and unvalidated.

Render: Blender; mannequin Mixamo/Adobe via three.js examples; HDRI/texture Poly Haven (CC0).

What we'd build: one cable set, three modes



One cable set

Slack; nudge ≤ 150 N; passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total.

Dual-mode spools

Servo for nudges plus a spring-applied, power-off slip brake.

Designed not to tip

Statics: tip SF ≥ 1.5 up to 420 N; unvalidated. 80 kg, CG 0.40 m, outriggers.

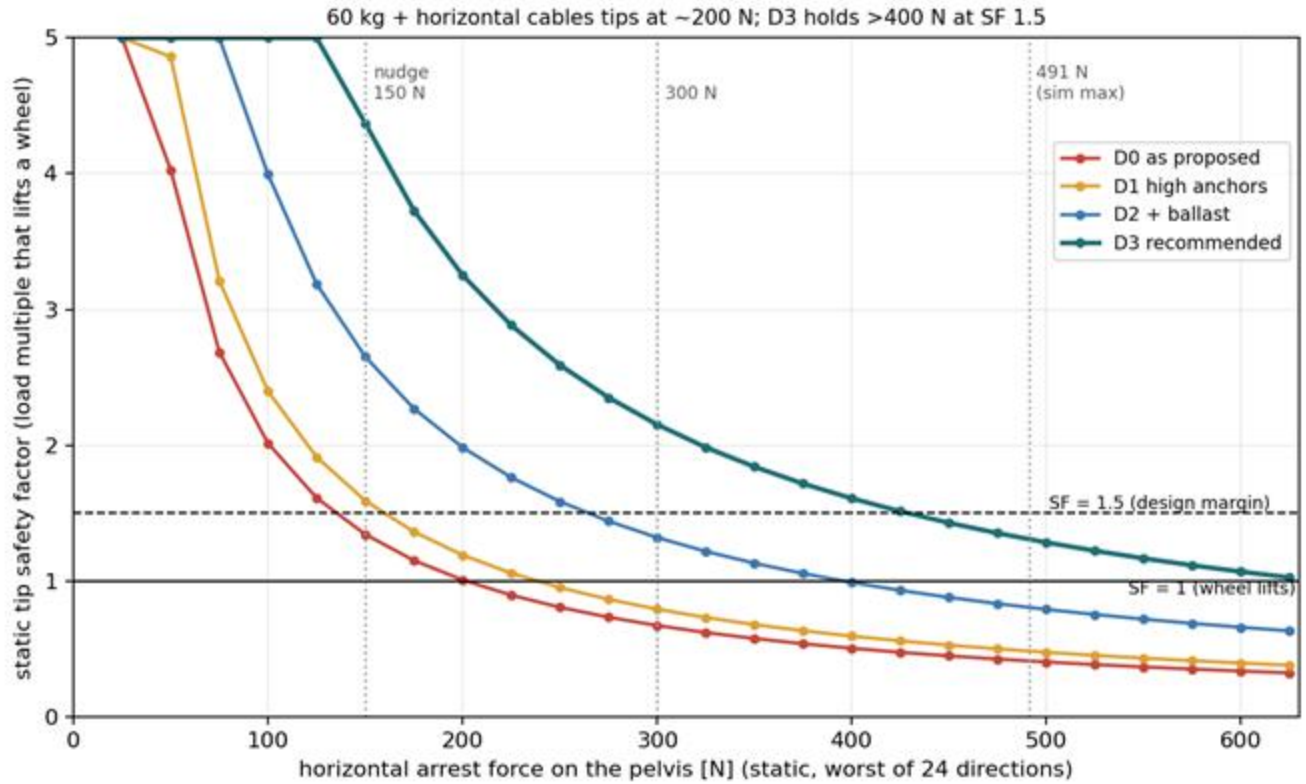
Concept, unbuilt

Statics and simulation only; not validated.

Render: Blender (design D3 CAD). Mannequin: Mixamo/Adobe via three.js examples; HDRI/texture: Poly Haven (CC0). Fits an existing walker, e.g. the NUS walker in Prof. Yu's lab (to be confirmed).

Can a walker catch without tipping?

In statics and simulation it is not predicted to tip, given an 80 kg low-CG walker, outriggers and mixed-height anchors.



805/814

caught trials arrested by the passive slip brakes (lock 0.28 s): 98.8% (Wilson CI 97.9–99.5%)

377/814

same trials with the original 60 kg design, fixed 300 N (46.3%)

380 ≤ 420 N

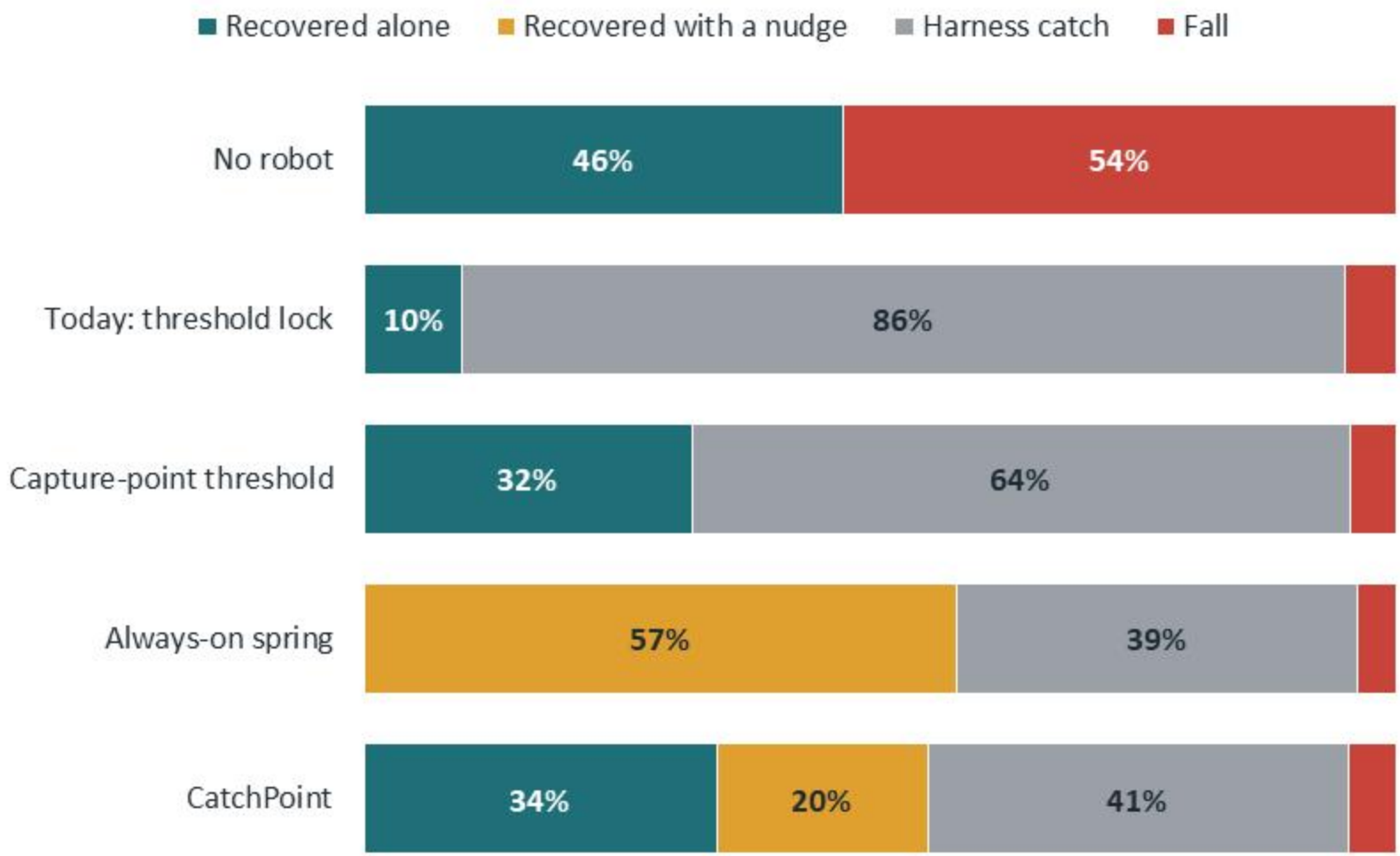
slip-resultant bound vs static tip envelope (SF 1.5); 360 motion directions

Coverage uses the force the passive slip brakes actually produce for each trial's fall direction: the along-fall component in the stopping model, the full resultant in the static-tip, slide and dynamic-tip checks. Brakes: passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total.

Statics + simulation only; not validated. The other 1186/2000 trials never engage the catch; this is not a fall-reduction claim.

Source: analysis/i2_sizing (report.md, data/results.json dyn.passive; CP m=0.03, lock 0.28 s, one seeded 2000-trial run). Static tip safety factor, worst of 24 directions.

At similar fall rates, far fewer harness catches



Ended without the harness

54.6% vs 9.6%

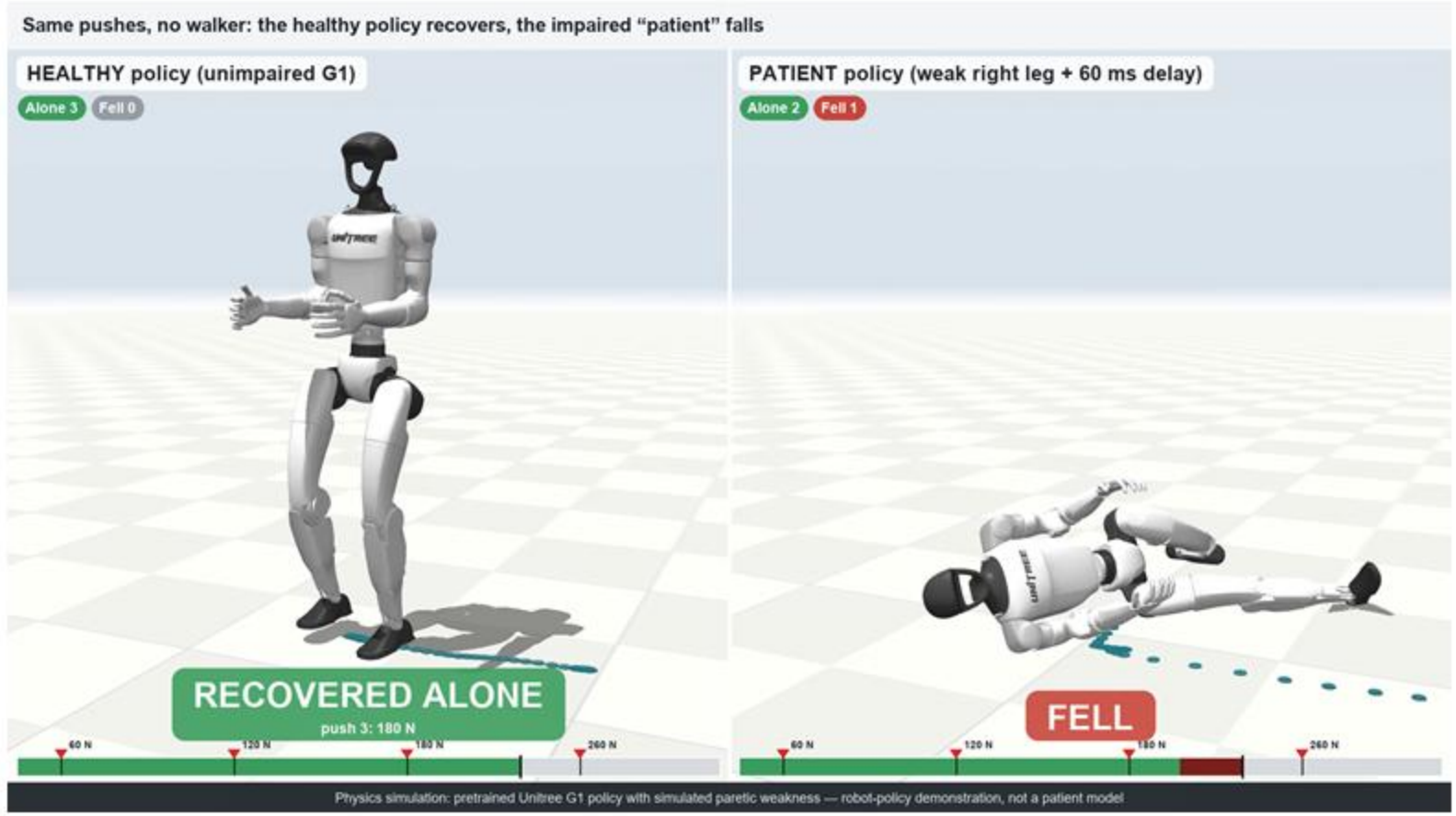
34.2% alone + 20.4% nudged, versus 9.6% for threshold lock, at ~5% falls (CatchPoint 4.8%, lock 5.0%).

Capture-point threshold: similar solo recoveries (31.8%) but harness in 63.7% vs 40.7%. 95% CIs up to ±4 pp. Simplified model, not clinical data.

Try it: interactive demo ↗

Source: results/summary.csv, ci.csv (seed 5412, n = 2000 pushes per row; selected operating points: B1 0.05 m, B1-XCoM 0.08 m, B2 200 N/m, CatchPoint margin 0.03).

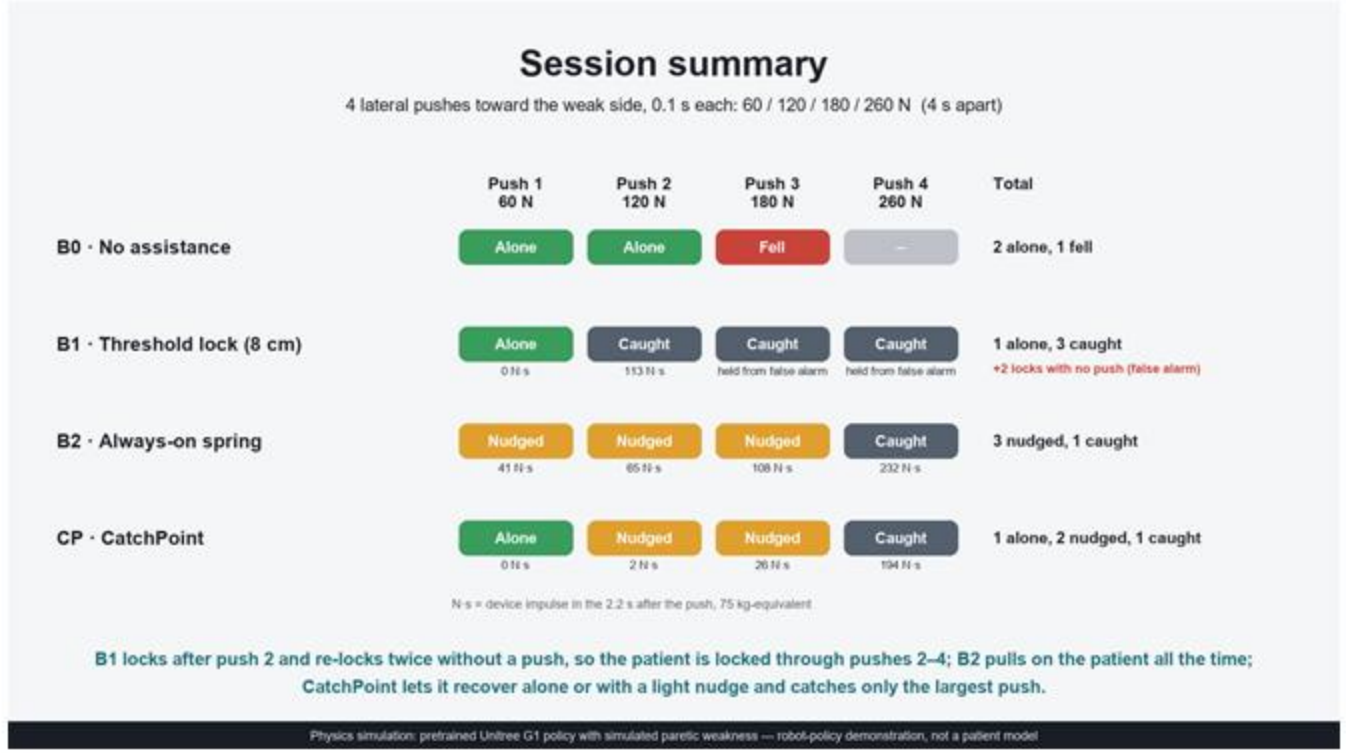
The physics patient: same pushes, healthy recovers, impaired falls



Pretrained Unitree G1 walking policy in MuJoCo. Patient = right-leg torque ×0.5 + 60 ms delay. Four lateral pushes toward the weak side (60 / 120 / 180 / 260 N, 4 s apart): the healthy policy recovers from all four; the patient falls at 180 N.

Source: physics_demo/videos_v2/open_ground_healthy_vs_patient.mp4 (one deterministic run), physics_demo/README.md v2. Robot-policy demonstration only — not a patient model.

Physics demo: fewer catch-locks, more low-force nudges



- MuJoCo demonstration using an impaired Unitree G1 policy: right-leg torque ×0.5, 60 ms delay, and 56 deterministic weak-side lateral pushes.
- Default calibrated CP: 0/56 falls, 26 catch-lock engagements, 23 nudges, 7 hands-off; B1 (θ=0.08 m): 0/56 falls, 46 catch-lock engagements, 10 hands-off; no robot: 32/56 falls.
- This demonstrates a simulated control trade-off; CP's recoverability region and escalation were calibrated on the same push grid. Image: one 4-push session (single gait phase; B1's re-locks partly come from the release rule).

Robot-policy demonstration only — not patient evidence, clinical validation, or a held-out performance test.

Source: physics_demo/sweep_all.json, README.md; videos: physics_demo/videos_v2/walker_4ways.mp4 (session summary image), physics_demo/videos/compare_b.mp4. Unitree G1 policy: unitree_rl_gym (BSD-3).

Why it makes a strong course project

I1 · core

New approach

Help only when recovery is no longer feasible

I1 · core

New control algorithm

Predict, check reach, push the minimum

I2 · feasibility

New mechanical structure

Walker-mounted cable interface, outriggers

I2 · feasibility

New actuation

Dual-mode spools: servo nudge + slip brake

supporting

Sensing fusion

Foot LiDAR, cable-length pelvis, load cells

no hardware needed

Demo-able

Simulation, CAD, interactive demo

Risks

Model realism. Simplified balance model: trends only.

Novelty. To our knowledge (search not exhaustive), no prior work combines prediction, per-patient step reach and a graded minimum nudge.

I2 unvalidated. Statics and simulation only; unbuilt.

Categories follow the course project briefing. I1 = control study, I2 = cable-interface feasibility. Novelty check: research/novelty_and_sources.md (search-based, not proof).

How the five of us could split it

Why

A · Clinical

What goes wrong for patients today, and how would we know we helped?

Problem, gap table, outcome measures, claim audit

Body

B · Mechanical

Can the walker nudge and catch without tipping?

D3 CAD, free-body diagrams, tip-margin table

Eyes

C · Sensing

How does the robot know the state fast enough?

Sensor layout, delay budget, demo video

Brain

D · Control

When does it act, and how much?

L0/L1/L2 flow, minimum-force rule, timing

Proof

E · Simulation

Does it beat today's approach?

Frozen Monte Carlo, CIs, ablations

Thu 22–Sat 24 Oct — Scope frozen, contracts agreed

Sun 25–Wed 28 Oct — Build slides and analysis

Thu 29 Oct — Merged draft deck

Fri 30–Sat 31 Oct — Claim audit, demo video

Sun 1–Mon 2 Nov — Two timed rehearsals

Tue 3 Nov — Submit slides

Source: docs/TEAM_PLAN.md. Each member owns one question and one talk segment. Dates: 2026.

It plugs into your ideas too



Illustration: team figure. Icons: Tabler Icons (MIT).

Decisions for this week

1

Confirm the scope

I1 control = core claim; I2 cable interface = feasibility study; I3 learner = appendix only.

2

Confirm who owns which role

Why / Body / Eyes / Brain / Proof — one question and one talk segment each.

3

Agree the interface contracts by Sat 24 Oct

Force limits, delay budget, D3 geometry and the frozen numbers file.

Next step: draft slides from all five by Thu 29 Oct; submit Tue 3 Nov.

Appendix

Technical depth for anyone who wants it

Architecture · Control law · Simulation model · Results · Robustness · I2 cable interface · Prior work · Other problems · I3 learner · References

A1 · Four layers: sense, estimate, decide, act

Sense

Cable lengths: pelvis pose

Shin-height 2D LiDAR: both feet

Load cells, IMU, BWS encoder

Estimate

COM state with Kalman filter

Capture point $\xi = x + \dot{x}/\omega$; base of support

Predict ahead over the delay

5-min patient calibration

Decide

Recoverable? Yes: L0 transparent

Smallest force ≤ 150 N: L1 nudge

Cannot restore: L2 slip-limited catch

Act

8 dual-mode spools: servo + slip brake

Overhead BWS cable, spool brake

Outriggers deploy on catch

Levels: L0 transparent · L1 nudge · L2 catch

Source: SPEC §3. Layering is our own sketch; sensor choices are proposals, not tested hardware.

A2 · Control law: predict, check, push no more than needed

Where must the foot land? (capture point)

$$\xi = x + \dot{x} / \omega$$

$\omega = \sqrt{(g / z_0)}$; x, $\dot{x}$ = centre-of-mass position and velocity.

Where will it end up with push F?

$$\xi_{land}(F) = p + (\xi - p)\, e^{\omega T} + \beta F$$

ξ = predicted capture point; p = centre of pressure; T = time to foot landing; $\beta = (e^{\wedge}\omega T - 1)/(m\omega^2)$.

Smallest push that keeps the step reachable

$$F^* = \operatorname{argmin} |F|^2 \quad \text{s.t.} \quad h(F) = r - |\xi_{land}(F) - a| \geq 0$$

r = the patient's step reach minus a margin; a = stance foot. Closed-form solution.

Which level?

- h(0) ≥ 0: patient recovers alone, F = 0
- |F*| ≤ 150 N: command F*
- |F*| > 150 N: command 150 N and catch

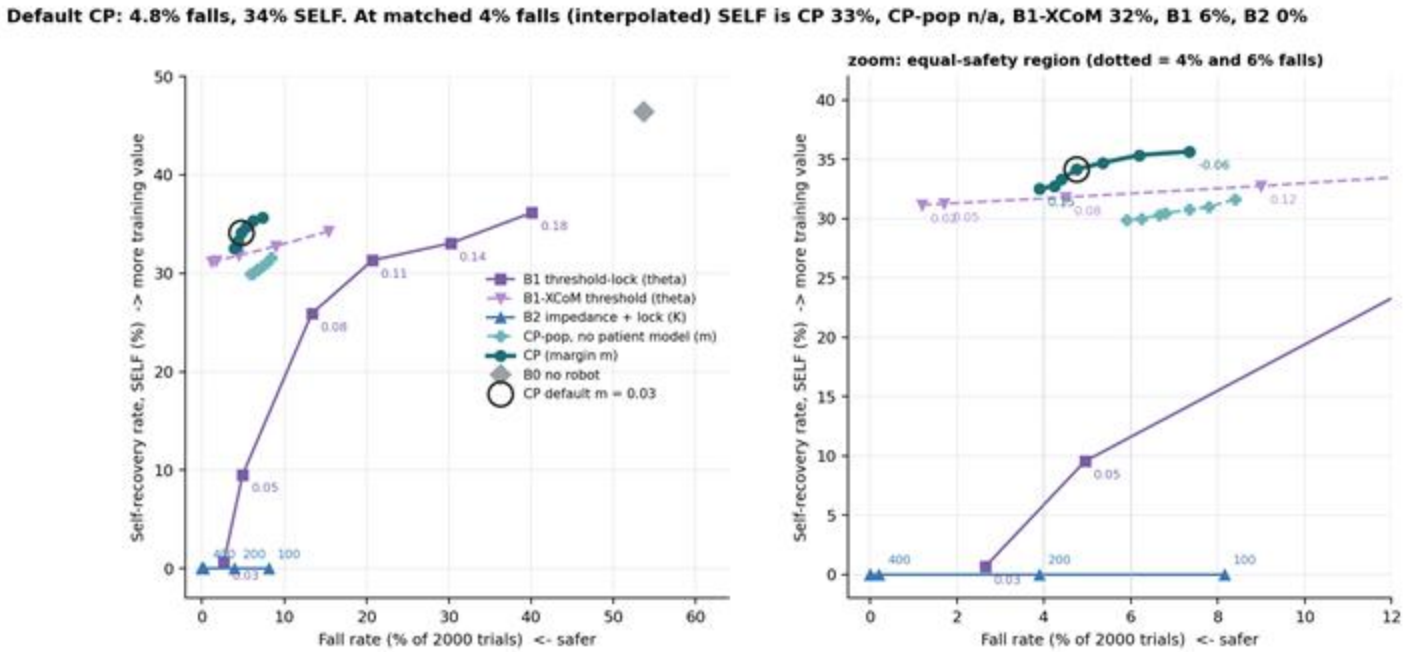
Sources: Hof et al., J Biomech 2005; Pratt et al., Humanoids 2006; control law as specified in SPEC §5.4 (our own design).

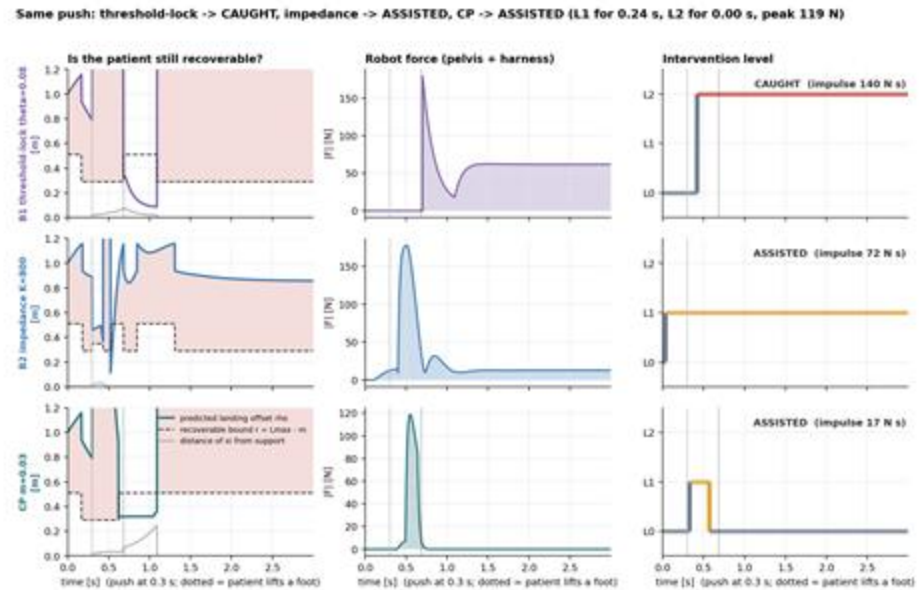
A3 · Simulation model and virtual patients

Parameter	Value or range	Note
Balance model	3D linear inverted pendulum	Horizontal plane, 2 ms step
Trial	3.0 s; push at 0.3 s	Δv 0.2–0.9 m/s, any direction
Population	50 patients × 40 pushes	2000 trials; same set for every controller (seed 5412)
Severity s	uniform 0–1	Drives all patient limits below
Mass / COM height	60–85 kg / 0.85–1.00 m	Sampled per patient
Reaction time	0.20 + 0.30·s	Seconds
Max step (paretic / other)	0.50 − 0.30·s / 0.60 − 0.10·s	Metres
Swing time (paretic / other)	0.40 + 0.25·s / 0.35 + 0.10·s	Seconds
Robot delays	sensing 30 ms; actuation 70 ms + 20 ms lag	Catch lock delay 0.28 s
Limits	Assist ≤ 150 N; fall = COM 0.25 m beyond support	Catch: harness spring-damper

All ranges are modelling assumptions anchored to literature, not clinical data (SPEC §5.2). s = severity.

A5 · More practice kept at the same fall rate (1/2)





Each controller traces a trade-off between falls and self-recovery as its setting is swept.

One showcase push: CatchPoint waits, nudges, and escalates only when needed.

Source: results/figs/pareto.png and timeline.png (project simulation, seed 5412).

A5 · Fair comparison at matched fall rates (2/2)

Solo recoveries, interpolated at equal falls

Controller	Solo recoveries at 4% falls	at 6% falls
Threshold lock (B1)	5.9%	11.6%
Capture-point threshold	31.7%	32.1%
Always-on spring (B2)	0.0%	0.0%
CatchPoint (CP)	32.6%	35.2%
CP, population model	n/a (not bracketed)	29.9%

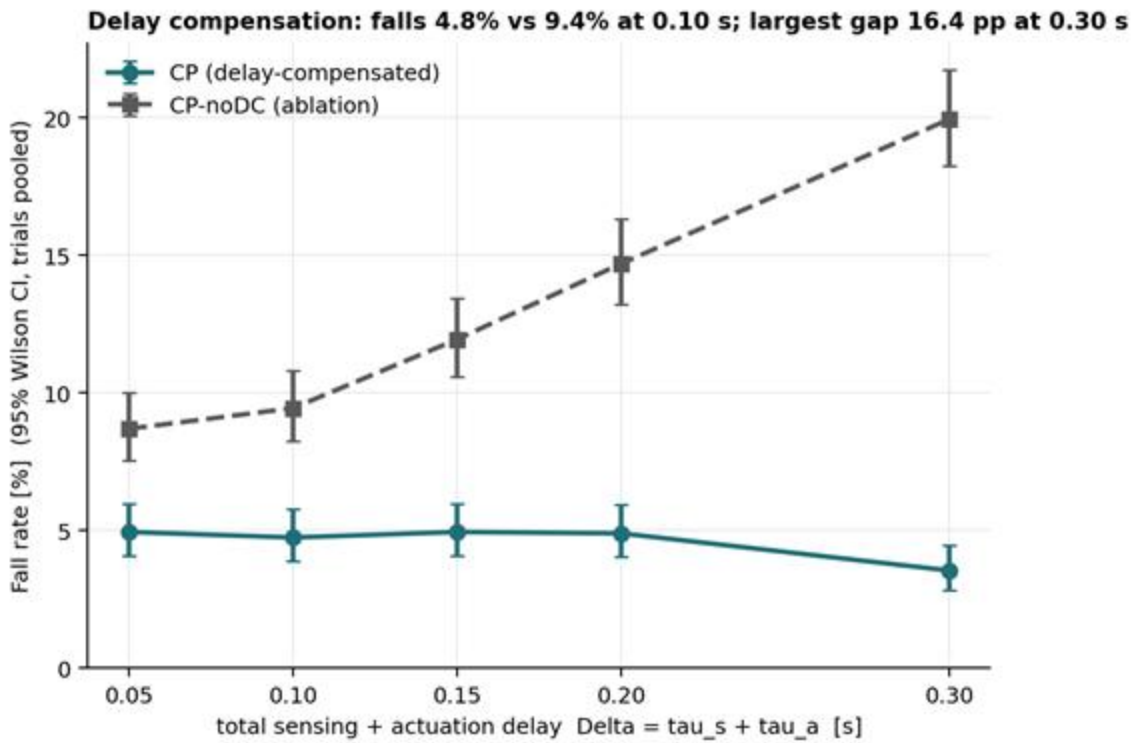
95% patient-bootstrap confidence intervals

Setting	Falls [95% CI]	Solo [95% CI]
Threshold lock 0.05	5.0% [3.4–6.6]	9.6% [7.2–11.9]
Spring 200 N/m	3.9% [2.9–4.9]	0.0% [0.0–0.0]
CatchPoint 0.03	4.8% [3.4–6.2]	34.1% [31.3–37.0]
CatchPoint 0.15	3.9% [2.7–5.3]	32.5% [29.8–35.0]

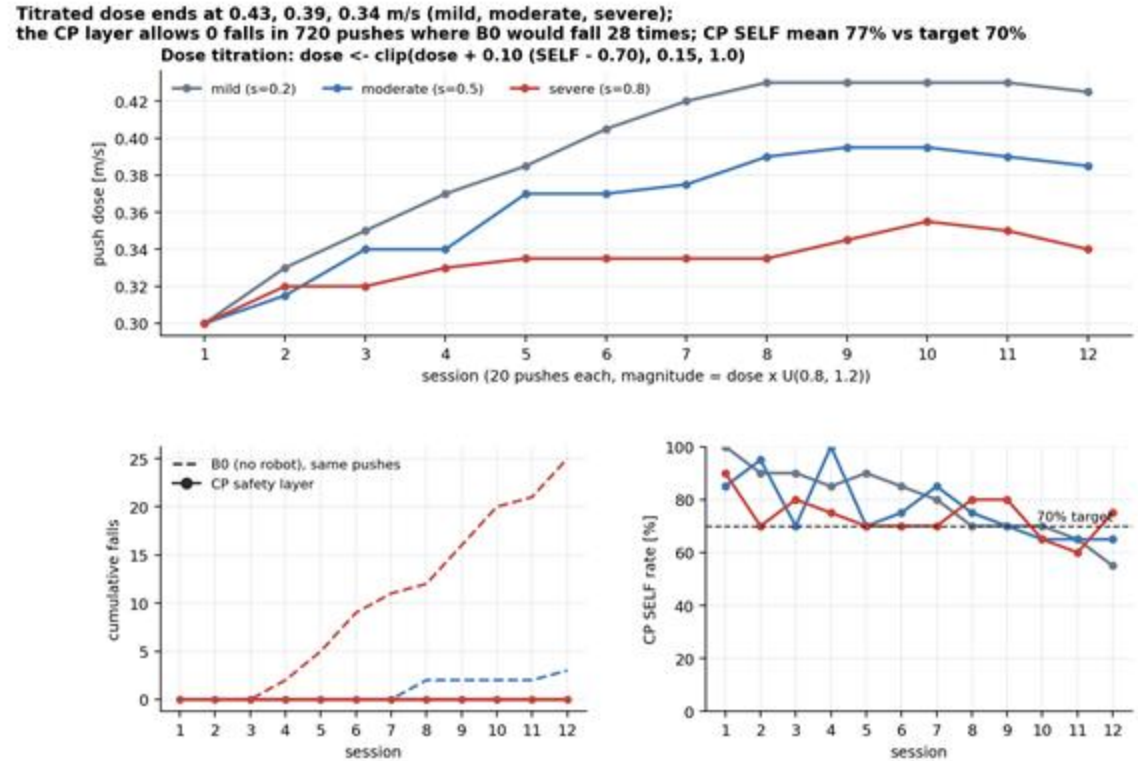
Ablation: the patient-specific model (CP) versus a population-average one (CP-pop, margin 0.03) is worth about 2 pp fewer falls and 4 pp more solo recoveries.

Source: results/matched.csv, ci.csv, summary.csv (simulation, seed 5412). Interpolated between swept settings; CI table text at 12 pt.

A4 · Prediction matters as delay grows (1/2)



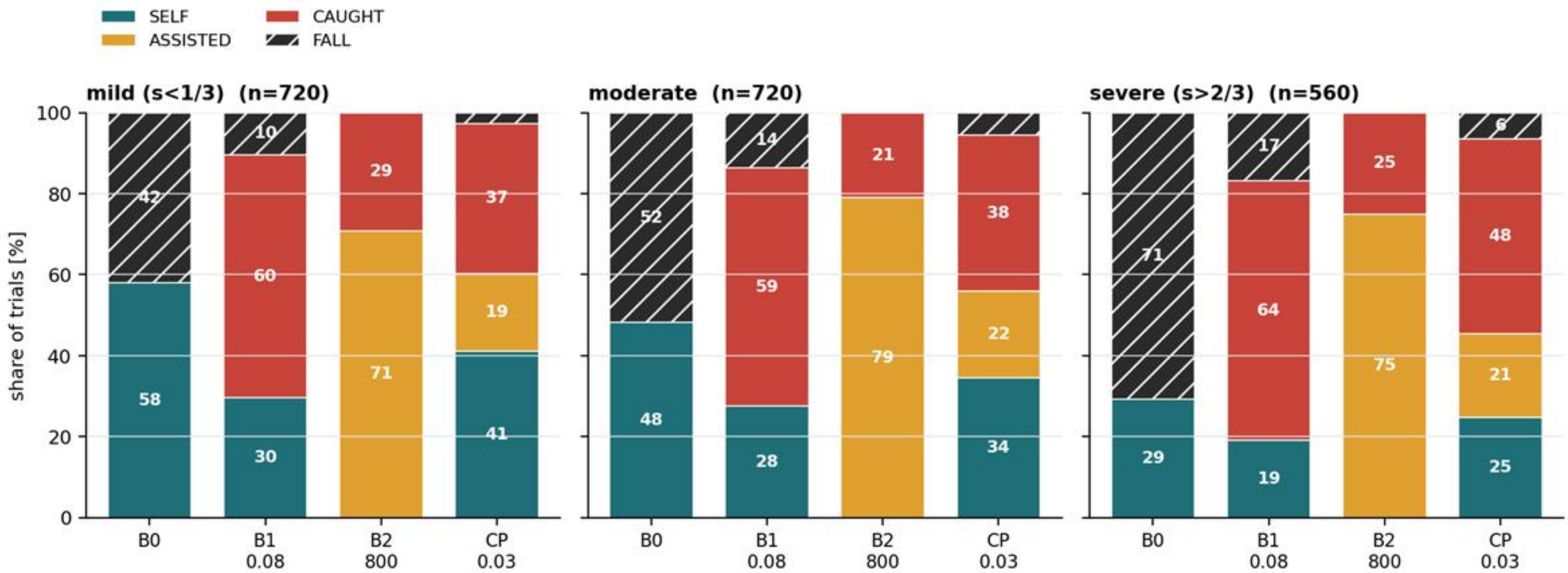
Without delay compensation falls rise with latency; with it they stay flat.



Adaptive dose: push size is tuned over 12 sessions toward a 70% self-recovery target.

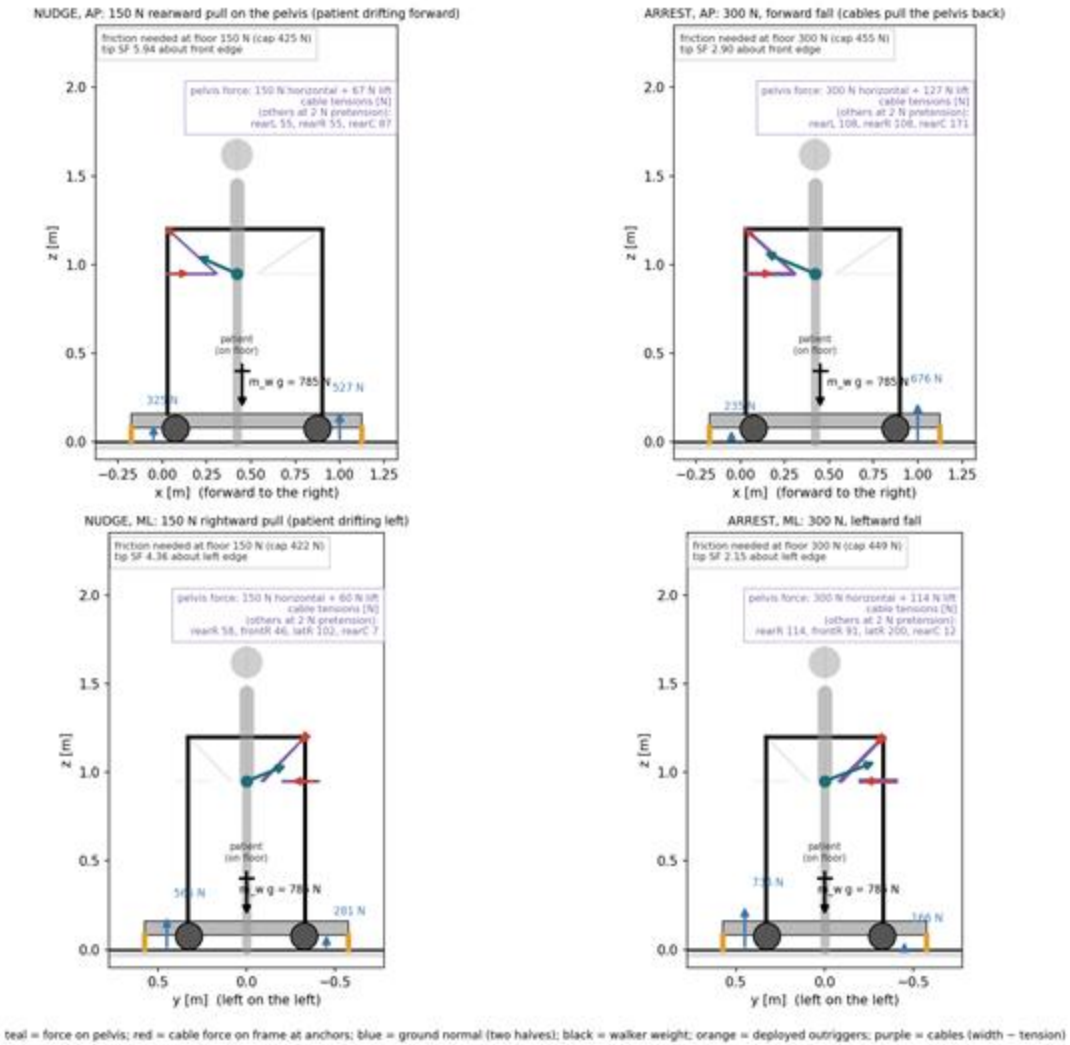
Source: results/figs/delay.png and aan_sessions.png (simulation).

A5 · Severe patients: CatchPoint falls 6%, keeps 25% self (2/2)



Source: results/figs/severity.png (simulation). B0 none, B1 threshold lock 0.08, B2 spring 800, CP = CatchPoint margin 0.03.

A6 · Free-body diagrams: nudge and arrest



Nudge, 150 N: tip safety factor 5.94 (AP) and 4.36 (ML); lift on the pelvis ≤ 66.1 N.

Arrest, 300 N, controller-set tensions: tip safety factor 2.90 (AP) and 2.15 (ML); friction needed 300 N against a cap of 455 N (AP).

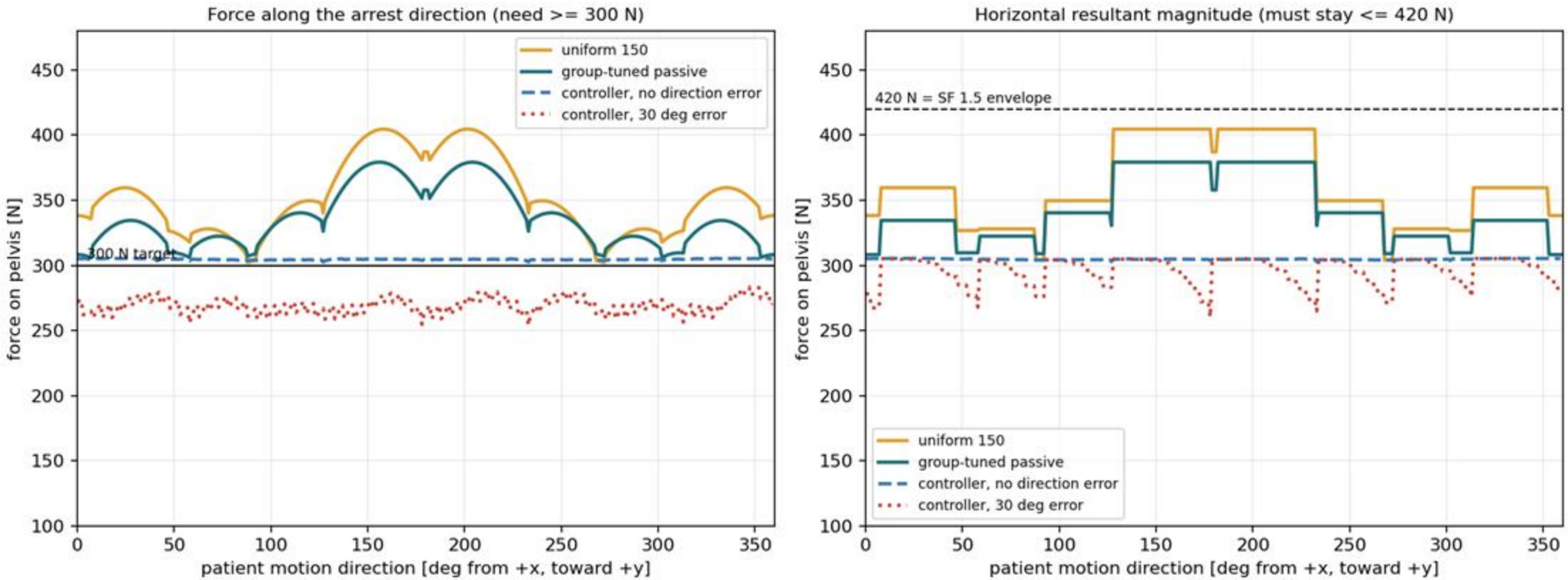
Passive slip brakes (A7): ≥ 300 N along the fall, ≤ 380 N total; tip SF ≥ 2.09 and sliding SF ≥ 1.28 at the full resultant.

Pelvis lift: 110.4–126.1 N at 300 N (≤ 171 N under passive slip), below the 294 N cap (0.5 body weight of 60 kg).

Statics only; outriggers deployed, mu = 0.5, rigid frame.

Source: analysis/i2_sizing/figures/fig6_fbd.png and data/results.json (arrest_rows, nudge_rows). Unbuilt and unvalidated.

A7 · Passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total



Group-tuned thresholds (corner 150, lateral 155, centre 120 N): ≥ 300 N along the fall (305 N minimum over 360 motion directions, nominal pelvis), worst-case horizontal resultant 380 N, below the 420 N static envelope (SF 1.5; SF 1.0 at 640 N). The perpendicular component reaches 178 N (28° off the fall); independent slipping does not preserve force direction. Tip SF ≥ 2.09, sliding SF ≥ 1.28 at the full resultant; the walker slides at 465 N (259 N if mu = 0.3).

Source: analysis/i2_sizing/data/slip.json (group_tuned), d3_contract.json, figures/fig7_slip_resultant.png. Unbuilt and unvalidated.

A8 · Arrest coverage and lock-delay sensitivity

Lock delay	Caught trials	D3, passive slip brakes (headline)	D3, fixed 300 N along the fall (old metric)	Original design, fixed 300 N
0.28 s (SPEC)	814	805/814 = 98.8% (CI 97.9–99.5)	804/814 = 98.7% (CI 97.7–99.4)	377/814 = 46.3% (CI 42.9–49.8)
0.15 s	844	821/844 = 97.2% (CI 95.9–98.2)	820/844 = 97.1% (CI 95.8–98.1)	170/844 = 20.1% (CI 17.5–23.0)
0.08 s	866	817/866 = 94.3% (CI 92.5–95.7)	807/866 = 93.1% (CI 91.3–94.7)	56/866 = 6.4% (CI 5.0–8.4)

Headline coverage uses the force the passive slip brakes actually produce for each trial's fall direction: the along-fall component in the stopping model, the full resultant in the static-tip, slide and dynamic-tip checks. Arrested = within 0.25 m, no dynamic tip, static tip SF ≥ 1.5, slide ≤ 1 cm. Wilson 95% CI.

Old metric, lock 0.28 s, fixed along-fall force of 200 / 300 / 400 N: 752 / 804 / 806 of 814. Allowing 491 N is worse: 775/814 = 95.2%, because the static SF-1.5 check fails above 420 N.

Baseline falls 95/2000; the 9 non-arrested caught trials could add to that: 104/2000 at worst. Not a fall-reduction claim.

COM stopped a median 0.104 s after engagement (p95 0.19 s; descriptive, nearest), 814 trials at lock 0.28 s. Slip-brake energy ≤ 22.2 J.

Source: analysis/i2_sizing/data/results.json (dyn.passive, dyn.curves). One seeded 2000-trial run; empirical and conditional on the D3 assumptions.

A9 · D3 contract: what the CAD must build

Item	Value
Arrest force (horizontal)	passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total (305 N minimum over 360 directions)
Static tip	SF ≥ 1.5 required; envelope 420 N (SF 1.5), 640 N (SF 1.0); SF ≥ 2.09 at the full slip resultant
Walker mass and CG	80 kg incl. ~20 kg low ballast; CG (0.45, 0, 0.4) m
Wheel contacts	x = 0.075 / 0.875, y = ±0.325 m (footprint 0.95 × 0.8 m)
Outrigger feet	+0.25 m all round, deployed on catch; rubber pads, mu ≥ 0.5
Anchors = cable exit points	4 corner at z = 1.20 m; 2 lateral and 2 centre at z = 0.95 m
BWS anchor (cantilever)	(0.42, 0, 1.8) m above the pelvis; hang load 1251 N
Nominal pelvis and belt	pelvis (0.42, 0, 0.95) m; belt half-depth 0.13 m, half-width 0.205 m
Open issue	Front-centre anchor post sits in the entry opening: needs a fold-down post

Geometry is read from the D3 contract (single source shared by CAD and analysis; the generated CAD parts are checked against it to 1 mm).

Source: analysis/i2_sizing/data/d3_contract.json and report.md section 8 (SPEC coordinates: x forward, y left, z up). Older cad/build_cad.py is the baseline SEA design, not D3.

A10 · Spool, brake and servo table

Quantity	Value	Basis
Passive slip thresholds	corner 150 N, lateral 155 N, centre 120 N	Spring-applied, power-off brakes
Slip torque (at the spool)	3.75 / 3.88 / 3.00 N·m	Threshold × spool radius 25 mm
Rated hold torque	5.63 / 5.82 / 4.50 N·m	1.5 × slip torque
Worst cable tension, 300 N arrest	199.6 N	Controller-set case, worst single cable, 24 directions
Nudge	150 N resultant, T_max 101.1 N per cable	Servo torque 2.53 N·m at the spool; tip SF ≥ 4.36
BWS spool brake	46.91 N·m; hang load 1251 N	1.5 × 85 kg × g, overhead, no horizontal load
Slip-brake energy	≤ 22.2 J (16.7 J at 0.28 s)	Passive slip, arrested trials, three lock delays
Pelvis lift caps	arrest 294 N (≤ 171 N under passive slip); nudge 88 N	0.5 BW and 0.15 BW of 60 kg

Not validated: no stress analysis, spool inertia or backlash; mu, cable stiffness and servo data are assumptions.

Source: analysis/i2_sizing/data/d3_contract.json, results.json (Tmax_arrest_worst, Tmax_nudge_worst, dyn.passive), slip.json. Replaces the earlier SEA sizing, which belonged to the baseline design.

A11 · Closest prior work and how CatchPoint differs

Work	What it does	How CatchPoint differs
RobUST (Omofuma 2023)	Cable pelvis robot; restoring force outside a fixed, subject-specific region	Predictive capture point, step-reach region, escalation to a catch
MRBA / DRBA (NTU)	Pelvic robots; sensor thresholds trigger a brake lock	Preserves the patient's own recovery step; graded force
ExoRecovery (Orhan 2023)	Exoskeleton plans and executes the recovery step	Patient keeps stepping; robot only helps
Monaco APO (2017)	Hip exoskeleton torque once a deviation is detected	Walker interface; recoverable-region logic
mTPAD (Stramel 2023)	Cable pelvic robot on a rollator, about 35 N per cable	Recoverability-based control; conceptual slip-limited catch mode

"To our knowledge": no work found that combines all four ingredients; absence of evidence from a search is not proof.

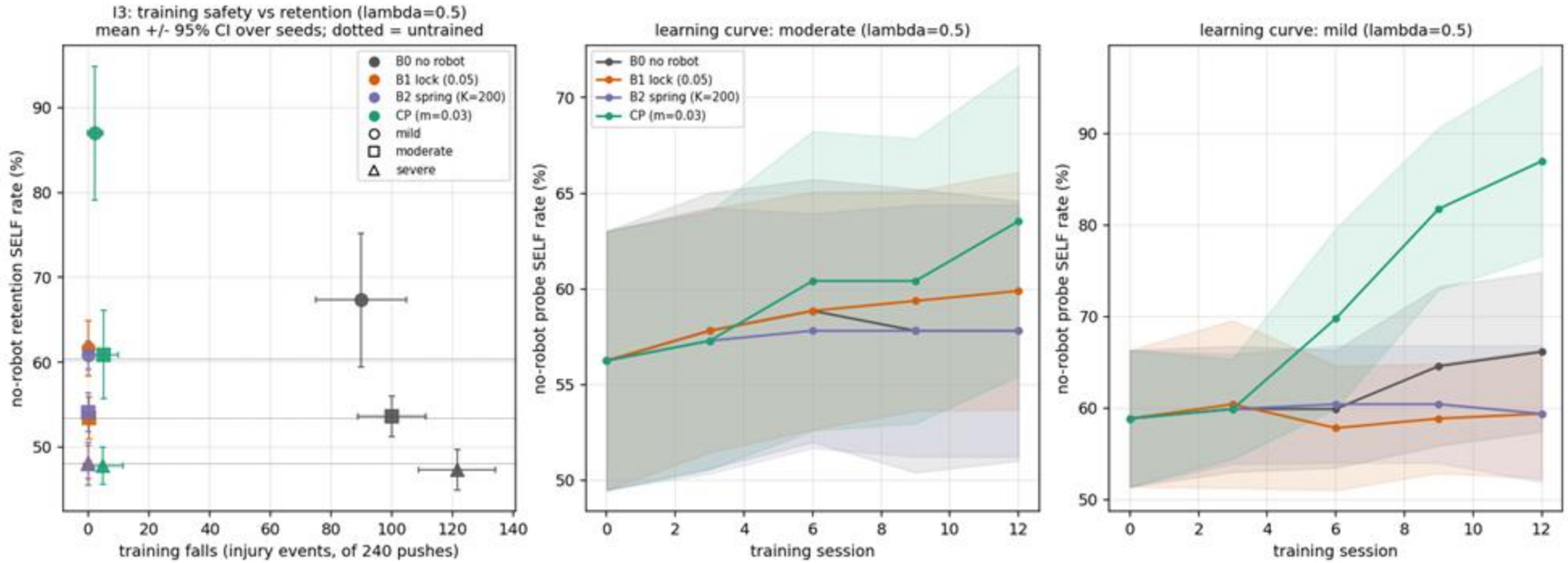
Sources: research/novelty_and_sources.md refs [1]–[5]. Some entries were read from abstracts or summaries only.

A12 · Other shortlisted problems

Idea	One-liner	Where “can they do it alone?” plugs in
P1 Sit-to-stand / transfer assist	Help people rise from a chair or bed	Centre of mass over the feet during the rise
P2 Upper-limb rehab	Robot-guided reaching practice	Trunk compensation instead of balance
P3 Dressing assistance	Arms that help put on clothing	Contact force with a moving limb
P4 Feeding assistance	Arm that brings food to the mouth	Contact force near the head

Source: SPEC §9 (mergeability). P1–P4 labels follow the team’s earlier shortlist; one-liners are brief paraphrases.

A13 · I3: surrogate-learner study (appendix only)



I3 is a four-parameter surrogate-learning study, not patient evidence. With one hand-tuned reward, 12 paired seeds per severity, 240 unique training pushes (1,200 candidate evaluations per learner) and 0.2–0.6 m/s disturbances, CatchPoint produced 2–5 training falls versus 90–121 for no robot, and improved no-robot retention (SELF) for mild (+19.7 ± 9.3 pp vs no robot) and moderate (+7.2 ± 6.1 pp) surrogate learners. The effect disappeared or reversed under the error-only reward ($\lambda = 0$), and severe learners did not improve. These results are properties of this surrogate, reward and disturbance dose; they do not establish patient learning or clinical benefit.

Source: results/i3_STATUS.md (wording), results/figs/i3_learning.png, results/i3_summary.md. Not part of the core claim.

A14 · Physics-demo protocol, calibration, and limits

Setting	Falls	Catch-lock	Nudged	Hands-off
No robot	32	–	–	24
B1 θ = 0.08 m	0	46	–	10
B1 θ = 0.11 m	0	41	–	15
B1 θ = 0.14 m	0	37	–	19
B1 θ = 0.18 m	1	31	–	24
CP r0 = 0.10 m	0	24	27	5
CP r0 = 0.14 m (default)	0	26	23	7
CP r0 = 0.20 m	0	33	14	9
CP r0 = 0.28 m	0	33	9	14

- 7 push magnitudes (60–300 N) × 8 gait phases; all pushes toward the weak side; forces mass-scaled to a 75 kg equivalent.
- B1 thresholds: θ=.08/.11/.14/.18 m. CP regions: r0=.10/.14/.20/.28 m. Both use the same delays, 0.28 s lock, cable model, and harness catch.
- PPO training used 9.6M steps of unassisted stochastic pushes; the same 56-grid test reduced B0 falls 32→3, with no held-out evaluation.

Kinematic walker; no cable/spool dynamics or collisions; harness torque is assumed; 'caught' denotes simulated catch-lock engagement.

Source: physics_demo/sweep_all.json, sweep_pareto.json, ppo/compare.json, README.md (deviations from SPEC: calibrated XCoM region, residual-based escalation).

A15 · References and image credits

[1] Mobile Robotic Balance Assistant (MRBA), J. NeuroEng. Rehabil., 2023.

[2] Y. Wang et al., "DRBA: Dynamic Robotic Balance Assistant," arXiv:2607.03027, 2026.

[3] Omofuma et al., "Training postural balance control with pelvic force field at the boundary of stability," Bioengineering, 2023.

[4] Monaco et al., "An ecologically-controlled exoskeleton can improve balance recovery after slippage," Sci. Rep., vol. 7, 2017.

[5] Z. O. Orhan et al., "ExoRecovery: Push recovery with a lower-limb exoskeleton," arXiv:2310.20339, 2023.

[6] D. M. Stramel et al., "Overground robotic gait trainer mTPAD improves gait symmetry and weight bearing in stroke survivors," Bioengineering, 2023.

[7] A. L. Hof et al., "The condition for dynamic stability," J. Biomech., 38(1):1–8, 2005.

[8] J. Pratt et al., "Capture point: A step toward humanoid push recovery," Humanoids, 2006.

[9] E. L. Inness et al., "Impaired reactive stepping among patients ready for discharge from inpatient stroke rehabilitation," Phys. Ther., 2014.

[10] A. Mansfield et al., "Is impaired control of reactive stepping related to falls during inpatient stroke rehabilitation?" Neurorehabil. Neural Repair, 2013.

[11] A. Mansfield et al., "Does perturbation-based balance training prevent falls?" Phys. Ther., 2015.

[12] Adeniyi et al., "Utilizing mobile robotics for pelvic perturbations… older adults," Sci. Rep., 2023.

[13] J. L. Emken et al., "Motor adaptation as a greedy optimization of error and effort," J. Neurophysiol., 97(6):3997–4006, 2007.

[14] V. C. V. Kumar et al., "Learning a control policy for fall prevention on an assistive walking device," arXiv:1909.10488, 2019.

[15] Patton et al., "KineAssist," Top. Stroke Rehabil., 15(2):131–139, 2008.

[16] World Health Organization, "Falls," fact sheet.

Images: US Navy photo (public domain) · NTU MRBA, CC BY 4.0 · RobUST, Omofuma et al., CC BY · Hocoma Andago, © Hocoma/DIH (fair use, educational) · Mannequin: Mixamo/Adobe via three.js examples · HDRI and textures: Poly Haven (CC0) · Icons: Tabler Icons (MIT) · Renders and figures: this project.